

Homework 4

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Problem 1.

(a) For the weather Markov chain with state order (S, C, R) and

$$A = \begin{bmatrix} 0.8 & 0.15 & 0.05 \\ 0.38 & 0.6 & 0.02 \\ 0.75 & 0.05 & 0.2 \end{bmatrix},$$

let $\pi = (\pi_S, \pi_C, \pi_R)$ be the stationary distribution. Solve $\pi = \pi A$, $\pi_S + \pi_C + \pi_R = 1$:

$$\pi = \left(\frac{638}{929}, \frac{245}{929}, \frac{46}{929} \right) \approx (0.6868, 0.2637, 0.0495).$$

Hence for a 4-day sequence $W_1 W_2 W_3 W_4$, its probability is $P(W_1 W_2 W_3 W_4) = \pi_{W_1} A_{W_1 W_2} A_{W_2 W_3} A_{W_3 W_4}$. The most probable sequence is

$$(S, S, S, S), \quad P(S, S, S, S) = \pi_S \cdot 0.8^3 \approx 0.3516.$$

(b) If today is rainy (R):

$$P(X_{t+1} = S \mid X_t = R) = A_{RS} = 0.75,$$

$$P(X_{t+2} = S \mid X_t = R) = (A^2)_{RS} = 0.75 \cdot 0.8 + 0.05 \cdot 0.38 + 0.2 \cdot 0.75 = 0.769.$$

(c) Let T_i be the expected number of days to reach sunny state S from state i . Then $T_S = 0$ and

$$T_C = 1 + 0.6T_C + 0.02T_R,$$

$$T_R = 1 + 0.2T_R + 0.05T_C.$$

Solving gives

$$T_R \approx 1.4107.$$

So starting from a rainy day, the expected waiting time to get a sunny day is about 1.41 days.

Problem 2.

1. For the Bayesian network with variables B (Battery), F (Fuel), G (Gauge), S (Start), the joint distribution decomposition is

$$P(B, F, G, S) = P(B)P(F)P(G \mid B, F)P(S \mid B, F).$$

2. Given $P(B = \text{bad}) = 0.1$, $P(F = \text{empty}) = 0.2$:

$$P(B = \text{good}, F = \text{empty}, G = \text{empty}, S = \text{yes}) = 0.9 \times 0.2 \times 0.8 \times 0.2 = 0.0288,$$

$$P(B = \text{bad}, F = \text{empty}, G = \text{not empty}, S = \text{no}) = 0.1 \times 0.2 \times 0.1 \times 1.0 = 0.0020.$$

Also,

$$\begin{aligned} P(S = \text{yes} \mid B = \text{bad}) &= \sum_{f \in \{\text{empty}, \text{not empty}\}} P(S = \text{yes} \mid B = \text{bad}, f)P(f) \\ &= 0 \cdot 0.2 + 0.1 \cdot 0.8 = 0.08. \end{aligned}$$

Problem 3.

1. For the Bayesian network with $M \rightarrow E$, and $(E, A) \rightarrow C$:

$$P(M, E, A, C) = P(M)P(A)P(E | M)P(C | E, A).$$

2. From the 40 training instances (MLE estimates):

$$\begin{aligned} P(M = Hi) &= \frac{20}{40} = 0.5, & P(M = Lo) &= 0.5, \\ P(A = Working) &= \frac{25}{40} = 0.625, & P(A = Broken) &= 0.375. \end{aligned}$$

$$\begin{aligned} P(E = Good | M = Hi) &= \frac{10}{20} = 0.5, & P(E = Bad | M = Hi) &= 0.5, \\ P(E = Good | M = Lo) &= \frac{15}{20} = 0.75, & P(E = Bad | M = Lo) &= 0.25. \end{aligned}$$

Integrating out M for $P(C | E, A)$:

$$\begin{aligned} P(C = Hi | E = Good, A = Working) &= \frac{3 + 9}{3 + 4 + 9 + 0} = \frac{12}{16} = 0.75, \\ P(C = Hi | E = Good, A = Broken) &= \frac{1 + 5}{1 + 2 + 5 + 1} = \frac{6}{9} = \frac{2}{3}, \\ P(C = Hi | E = Bad, A = Working) &= \frac{1 + 1}{1 + 5 + 1 + 2} = \frac{2}{9}, \\ P(C = Hi | E = Bad, A = Broken) &= \frac{0 + 0}{0 + 4 + 0 + 2} = 0. \end{aligned}$$

Therefore,

$$P(C = Lo | E, A) = 1 - P(C = Hi | E, A).$$

3. For $P(E = Bad, A = Broken)$:

$$\begin{aligned} P(E = Bad, A = Broken) &= P(A = Broken) \sum_m P(m)P(E = Bad | m) \\ &= 0.375 \times (0.5 \times 0.5 + 0.5 \times 0.25) = 0.140625. \end{aligned}$$